



Rapid synthesis of silver nanoparticles using dried medicinal plant of basil

Naheed Ahmad^a, Seema Sharma^{b,*}, Md. K. Alam^a, V.N. Singh^c, S.F. Shamsi^d, B.R. Mehta^c, Anjum Fatma^e

^a Department of Botany, Patna University, Patna 800013, India

^b Department of Physics, A.N. College, Magadh University, Patna 800013, India

^c Thin Film Laboratory, Department of Physics, IIT Delhi, New Delhi 110015, India

^d Department of Electrical & Electronics Engineering, SMIT, Sikkim, Gangtok, India

^e Department of Chemistry, Patna University, Patna 800013, India

ARTICLE INFO

Article history:

Received 5 June 2010

Received in revised form 23 June 2010

Accepted 24 June 2010

Available online 24 July 2010

Keywords:

XRD

Green synthesis

NPs

ABSTRACT

Plants respond to heavy metal stress by metal complexation process like production of phytochelations or by other metal chelating peptides. In this paper we report the synthesis of silver nanoparticles (AgNPs) from the room dried stem and root of *Ocimum sanctum*. The broth of the plant is used as a reducing agent for the synthesis of Ag nanoparticles at room temperature. The reaction process was simple and was monitored by ultraviolet–visible spectroscopy (UV–vis). There was formation of highly stable silver nanoparticles in the solution. The morphology and crystalline phase of the NPs were determined from transmission electron microscopy (TEM), selected area electron diffraction (SAED) and X-ray diffraction (XRD) spectra. Transmission Electron Microscopy studies showed that the silver nanoparticles obtained from roots and stem were of sizes 10 ± 2 and 5 ± 1.5 nm, respectively. The various phytochemicals present within the ocimum plant result in effective reduction of silver salts to nanoparticles but their chemical framework is also effective at wrapping around the nanoparticles to provide excellent robustness against agglomeration.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Noble metal nanoparticles (NPs) are well known to have important applications in the fields of electronic, magnetic, optoelectronics, and information storage [1–4]. It is now understood that the intrinsic properties of a noble metal NP are determined by its size, shape, composition, crystallinity, and structure (solid or hollow). Silver NPs, as a significant member of the noble metal NPs, are excellent substrates for Surface Enhanced Raman Scattering (SERS) [5] to probe single molecules, and are excellent as catalysts for accelerating some chemical reactions [6]. A number of approaches are available for the synthesis of silver NPs. For example, silver ions are reduced by chemical [7] electrochemical [8], radiation [9], photochemical methods [10], Langmuir–Blodgett [11,12] and biological techniques [13]. Among these methods, biological syntheses are not only a good way to fabricate benign nanostructure materials, but also to reduce the use or generation of hazardous substances to human health and the environment. This approach provides a facile and convenient entry to producing multiple inorganic NPs. People have used purified natural materials to prepare noble metal NPs, such as vitamin B2 to synthesize gold and

platinum NPs with interesting morphologies [14]. Vitamin C has been used to prepare gold NPs in molecular gels and liquid crystal systems [15]. Now, there is more and more interest in green synthesis research. Synthesis using bio-organisms, especially plants that secrete the functional molecules for the reaction, is compatible with the green chemistry principles: the bio-organism is (i) ecofriendly as are (ii) the reducing agents employed, and (iii) the capping agent in the reaction. There are some examples of synthesizing nanomaterials using plants, including the use of live alfalfa to synthesize Au and AgNPs [16], the secretions from germinating chickpea seeds to biosynthesize CaCO_3 crystals [17], and the broth of lemongrass and geranium leaves to synthesize Au and AgNPs [18,19]. Recent works provide evidence that some higher plants possess the capacity to absorb and sequester heavy metals [20]. Members of the Brassicaceae family as well as water hyacinth [21], were found to be high Cr accumulator. Cassava peels have been used for heavy metal reduction [22], *Thuja occidentalis* and *Tectona grandis* leaves were reported to reduce Cu [23,24].

Nanobiotechnology is a promising field especially for biodiversity rich countries like India. Biological diversity can be thus used as a major resource for biotechnological products and processes, which may be suitable for large scale synthesis. In this paper we report the synthesis of AgNPs from the sun dried stem and root of *basil plant*. *Ocimum sanctum* or Tulsi as it is commonly known in India is native throughout the old world tropics and widespread as a cultivated plant and an escaped weed. It is cultivated for religious

* Corresponding author.

E-mail address: seema.sharma26@yahoo.com (S. Sharma).

and medicinal purposes, and for its essential oil. The choice of the plant was motivated by the presence of aromatic compounds. The plant contains a variety of phytochemical compounds (such as phenols, amino acids, flavones etc.) and exhibit antioxidant activity. The plant extract are reported to have activities of scavenging superoxide anion radicals, 1,1-diphenyl-2-picrylhydrazyl radicals (DPPH), hydroxyl radicals, hydrogen peroxide, chelating ferrous iron, and ferric ion reducing potential. These molecules are expected to self assemble and cap the metal nanoparticles formed in their presence and thereby induce some shape control during metal ion reduction.

To the best of our knowledge, the use of ocimum plant extract at room temperature for the green synthesis of noble metal NPs, such as silver NPs, has not been reported. It is important to study the formation of NPs in ocimum root and stem extracts for the understanding of the mechanism of biological synthesis and for exploring new green ways to prefabricate nanomaterials. Thus, we report here the synthesis and antibacterial activity of Ag nanoparticles from the sun dried stem and root of *basil* plant.

2. Experimental details

2.1. Materials and methods

2.1.1. Synthesis of silver nanoparticles

Nanoparticles of Ag were prepared by using ocimum plant extract. Known weight of healthy plant/plant part was taken and washed thoroughly in running water for 20 min. This was followed by another rinse in sterile distilled water. Now plant samples were placed in a 250 ml beaker containing 100 ml of distilled water and were placed on boiling steam bath for 15–20 min till colour of the solvent changes. They were cooled at room temperature, gently pressed and filtered firstly through sterile serene cloth and through Whatman filter paper. This solution was treated as source extract and was utilized in subsequent procedures.

40 ml of source extract of plant was mixed with (10 ml) of 0.025 (M) AgNO_3 solution. The extracts were warmed at 40 °C on the steam bath for 10 min until the colour of solutions changes from light yellow to brown (Fig. 1c). They were allowed to cool in the laboratory ambience. The deposition gets distinctly visible in the flask which were left for 5 h and subsequently filtered. The retrieval of

AgNPs was simple and easy. The solution containing AgNPs which was reconfirmed by the signatory colour (dark brown) was then poured out into petri dishes and left in the oven for drying at 250 °C for 24 h. The water evaporated leaving behind the basil mediated AgNPs, which were scraped out for further study. The rate of nanoparticle synthesis was very high, which justifies use of plants over microorganisms in methods. This process for the production of silver nanoparticles uses the direct interaction of silver nitrate with basil plant (root and stem) extract in aqueous media without the intervention of any external man-made chemicals. Therefore, this reaction pathway satisfies all the conditions of a 100% green chemical process. Silver nanoparticles produced by this process did not require any external chemicals for the stabilization of the nanoparticulate matrix. Secondary metabolites present in the Ocimum plant are presumably responsible for the creation of a robust coating on silver nanoparticles and thus, rendering the nanoparticles stable against agglomeration.

2.1.2. Characterization

The bioreduction of Ag^+ ions with the two samples of stem and root was monitored by periodic sampling of the aliquots (1 ml) of the aqueous component. The optical absorbance of silver nanoparticles suspended in distilled water was recorded on UV–vis spectrophotometer (Systronics 2202 double beam model) in 200–800 nm wavelength range. The formation and quality of compounds were checked by XRD technique. The X-ray diffraction (XRD) pattern measurements of drop-coated films AgNPs on glass substrate were recorded in a wide range of Bragg angles 2θ at a scanning rate of 2° min^{-1} , carried out on a Philips PW 1830 instrument that was operated at a voltage of 40 kV and a current of 30 mA with Cu $K\alpha$ radiation (1.5405 Å). High Resolution Transmission Electron Microscopy (HRTEM) was performed by TECHNAI G20-STWIN (200 kV) machine with a line resolution 2.32 (in Å). These images were taken by drop coating AgNPs on a carbon-coated copper grid. Energy Dispersive Absorption Spectroscopy photograph of AgNPs were carried out by the HRTEM equipment as mentioned above.

3. Results

Fig. 1a displays the UV–vis spectra of stem solution as a function of reaction time. The strong resonance centered at 440 nm was clearly observed and increased in intensity with time. It might arise from the excitation of longitudinal plasmon vibrations in silver NPs in the solution. The results were in accordance with the report of Naik et al. [13]. The characteristic brown colour of silver solutions provided a convenient spectroscopic signature to indicate their formation. These colours arise due to excitation of the surface plasmon resonance in the (SPR) in the metal nanoparticles. It was observed that the plasmon intensity at the reaction time of 11 h is near to that at 15 h, meaning completion of the reaction. Ag nanoparticles were observed to be stable in solution and show very little aggregation. Besides, the plasmon bands are broadened with an absorption tail in the longer wavelengths, which may be due to the size distribution of the particles.

The roots showed slightly different pattern of SPR bands (at 442 nm) with relatively sharper plasmon peak than the stem. The peaks (Fig. 1b) are typical of the absorptions of proteins. The peak at 210 nm was assigned to the strong absorption of peptide bonds in the extract. The absorption at 280 nm indicated the presence of tyrosine, tryptophan, and/or phenylalanine residues in the proteins, which are known to interact with silver ions [25].

Transmission electron microscopy images of Ag nanoparticles are shown in Fig. 2a and b derived from basil plant root and stem, respectively. The silver nanoparticles obtained from roots and stems were of sizes 10 ± 2 and 5 ± 1.5 nm, respectively. The

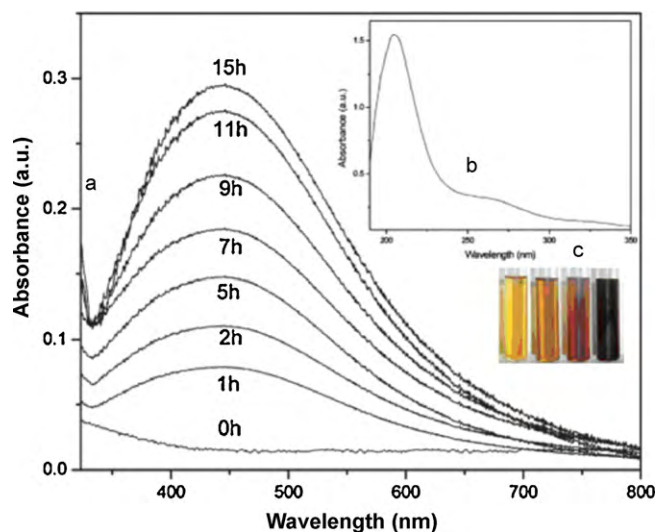


Fig. 1. UV/vis spectra recorded from the culture supernatant (derived from the root of the plant) that shows the production of AgNPs after various periods of time. (c) Photographs showing change in colour after adding AgNO_3 , before reaction and after reaction time of 1 h, 6 h and 11 h.

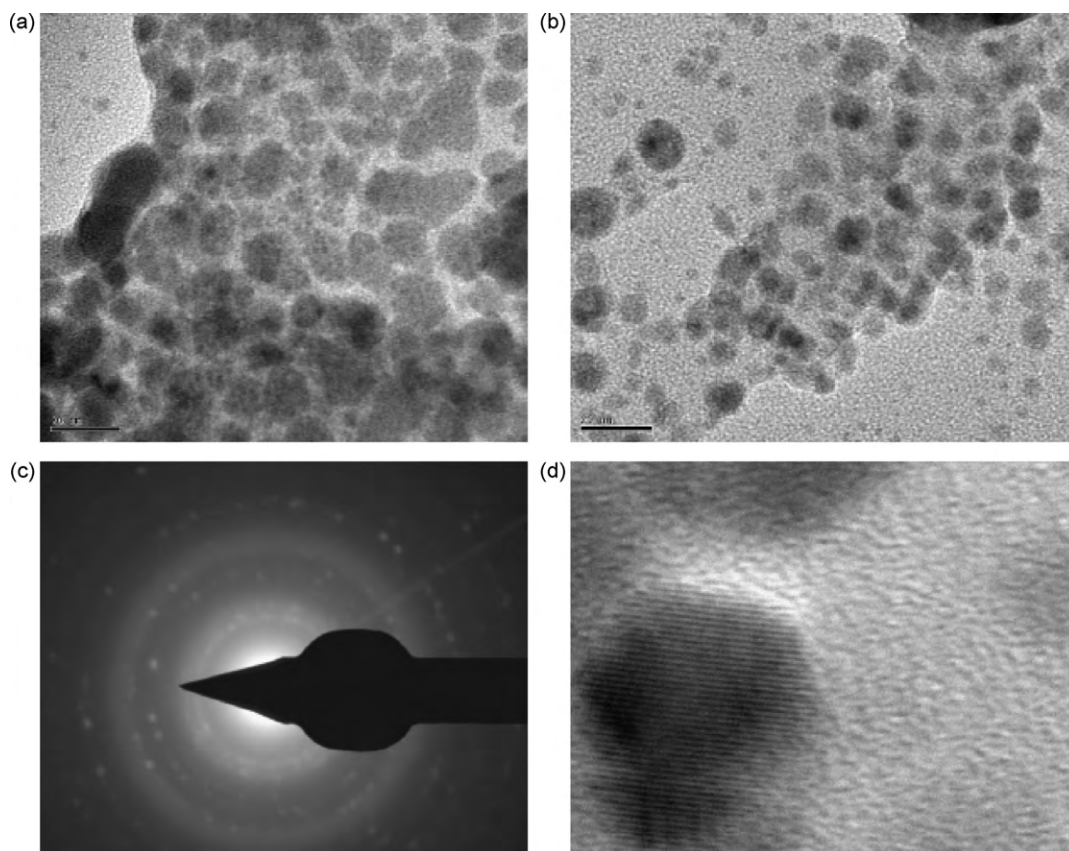


Fig. 2. Transmission electron microscopy images of AgNPs derived from (a) root and (b) stem. (c) Selected area electron diffraction showing the characteristic crystal planes of elemental silver. (d) HRTEM image showing characteristic d spacing for the $[200]$ plane.

morphology of the NPs was spherical. It was noticeable that the edges of the particles were lighter than the centers, suggesting that biomolecules, such as proteins in basil plant capped the silver NPs. It is seen that proteins are present among the particles and are adhered to their surfaces. Fig. 2c shows the SAED pattern of stem derived silver nanoparticles that revealed the diffraction rings from inner to outer, with d -spacings of 0.236, 0.2202, 0.141, and 0.121, could be indexed as (1 1 1), (2 0 0), (2 2 0), and (3 1 1) reflections, respectively, corresponding to face-centered cubic (fcc) silver. The diffraction rings also suggested the NPs were polycrystalline. HRTEM image (Fig. 2d) shows the d spacing of 2.02 Å, which matches with the $[200]$ plane of AgNPs. The morphology and crystalline phase of the silver NPs was not affected by the increase

in silver ions. Histograms (Fig. 3a and b) show NPs having an average size 10 ± 2 and 5 ± 1.5 nm derived by basil plant root and stem, respectively. The size of the nanoparticles derived from stem ($\sim 62\%$) is more uniform than that derived from the root ($\sim 30\%$).

Fig. 4 shows XRD patterns of silver (root) (a), and (stem) (b), respectively. The Bragg reflections show that the two samples were in the same space group. The peaks were assigned to diffraction from the (1 1 1), (2 0 0), (2 2 0), (3 1 1) and (2 2 2) planes of face-centered cubic (fcc) silver, which were in good agreement with reference to the unit cell of the fcc structure (JCPDS File No. 87-0720) with a lattice parameter of $a = 4.077$ Å. It is worth noting that the relative intensity of the (2 0 0) to (1 1 1) diffraction peaks in Fig. 4a and b (0.50, 0.54) was higher than the conventional value

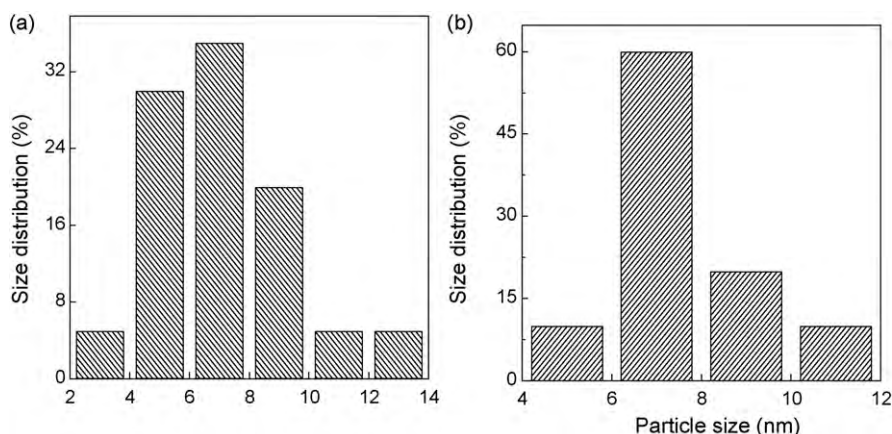


Fig. 3. Histogram showing the average sizes of AgNPs derived from (a) root and (b) stem.

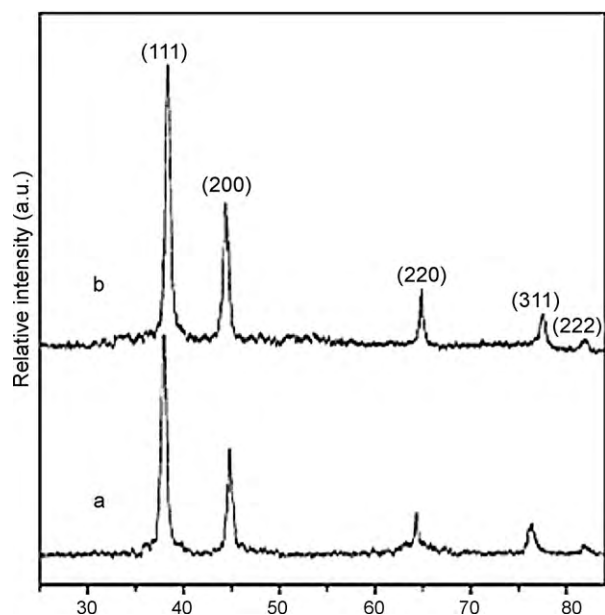


Fig. 4. X-ray diffractogram of silver nanoparticles synthesized by basil plant extract: (a) root derived and (b) stem derived.

(0.45). This result indicated that the synthesized silver NPs in our system were enriched in {1 0 0} facets, and thus their {1 0 0} planes tended to be preferentially oriented parallel to the surface of the supporting substrate [25]. In addition, the intensity ratio of the (2 0 0) to (1 1 1) peaks of silver (root) (0.54) was higher than for silver (stem) (0.50), which suggested that {1 0 0} planes were easier to grow in a higher concentration of the silver ion solution by allowing more silver atoms to accumulate onto them. Meanwhile, the size of these particles could be estimated from the Debye–Scherrer Eq. (1) by determining the width of the (1 1 1) Bragg reflection [26].

$$L = \frac{k\lambda}{\beta_{1/2} \cos \theta} \quad (1)$$

k is the Scherrer constant, λ is the wavelength of the X-ray, β and θ are the half width of the peak and half of the Bragg angle, respectively. Therefore the size (L) of the particles could be easily estimated; the sizes of silver NPs (stem) and (root) were about 11.5 and 7.9 nm, respectively. These results were consistent with the sizes of NPs obtained from the TEM study as illustrated above.

Fig. 5a and b shows the Energy Dispersive Absorption Spectroscopy photographs of root and stem derived AgNPs. All the peaks of Ag are observed and are assigned. Peaks for Cu and C are from the grid used and the peaks for S, P and N correspond to the protein capping over the AgNPs.

4. Discussion

A well-accepted scientific consensus emanating from several scientific investigations is that basil plant contains high levels of antioxidant polyphenols, including flavonoids. The phenolics and other phytochemicals within the ocimum plant not only result in effective reduction of silver salts to nanoparticles but their chemical framework is also effective at wrapping around the nanoparticles to provide excellent robustness against agglomeration. The current discovery of the unique chemical power of non-toxic phytochemicals in the plant initiating nanoparticle formation is of paramount importance in the context of the production of silver nanoparticles for medical and technological applications under non-toxic conditions. Phytochemical analysis of ocimum plant extract indicated the presence of major phytochemicals, includ-

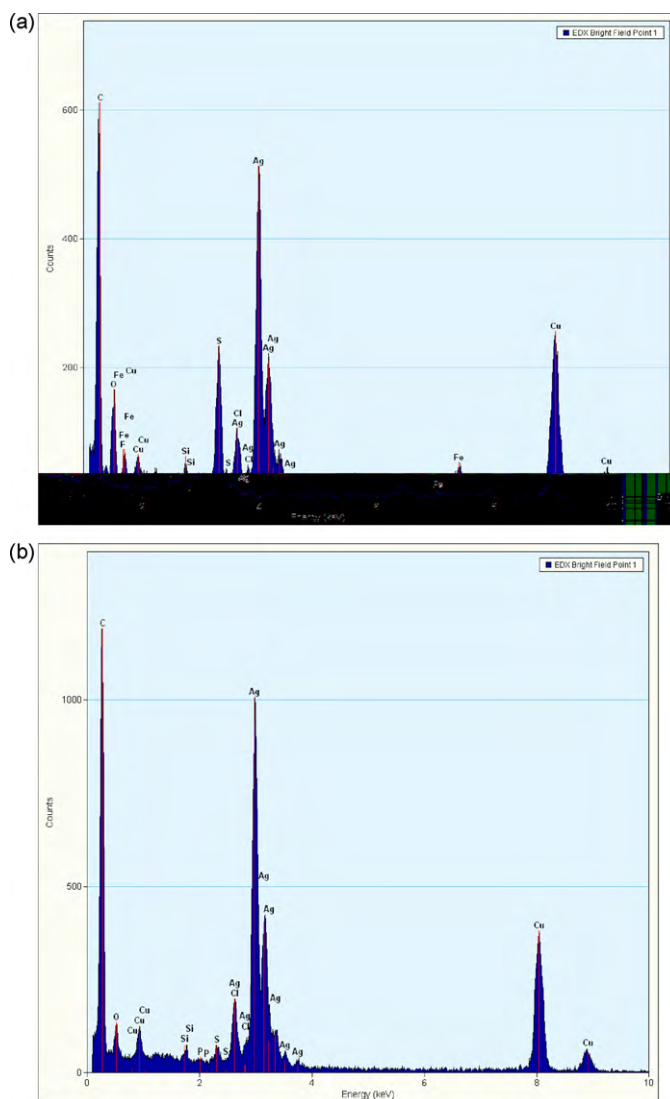


Fig. 5. Energy dispersive absorption spectroscopy photograph of AgNPs derived by (a) root and (b) stem.

ing phenolics, alkaloids, glycosides, flavonoids (Apigenin, Luteolin), and tannins. The phenolic concentrations have been found to be 80 mg/g in the dry plant extract and a fair correlation between antioxidant/free radical scavenging activity and phenolic content was observed [27]. The phenolic components in the plant extract are rosmarinic acid, lithospermic acid, vanilic acid, p-coumaric acid, hydroxybenzoic acid, syringic acid, caffeic acid, ferulic acid, cinnamic acid, dihydroxy phenyllactic acid, and sinapic acid. Rosmarinic acid and flavonoids like luteolin, apigenin, orientin, vicenin and isoeugenol [28] were the most abundant components identified in basil plant.

Phenolic compounds possess hydroxyl and carboxyl groups, which are able to bind to metals [29]. The roots of many plants contain high levels of phenolic compounds which may inactivate ions by chelating. According to Morgan et al. [30] this general chelating ability of phenolic compounds is probably related to the high nucleophilic character of the aromatic rings rather than to specific chelating groups within the molecule.

Flavonoids can also directly scavenge molecular species of active oxygen. Their antioxidant action resides mainly in their ability to donate electrons or hydrogen atoms.

Rosmarinic acid (alpha-o-caffeoyl-3,4-dihydroxyphenyl lactic acid) exhibits significant free radical scavenging, anti-

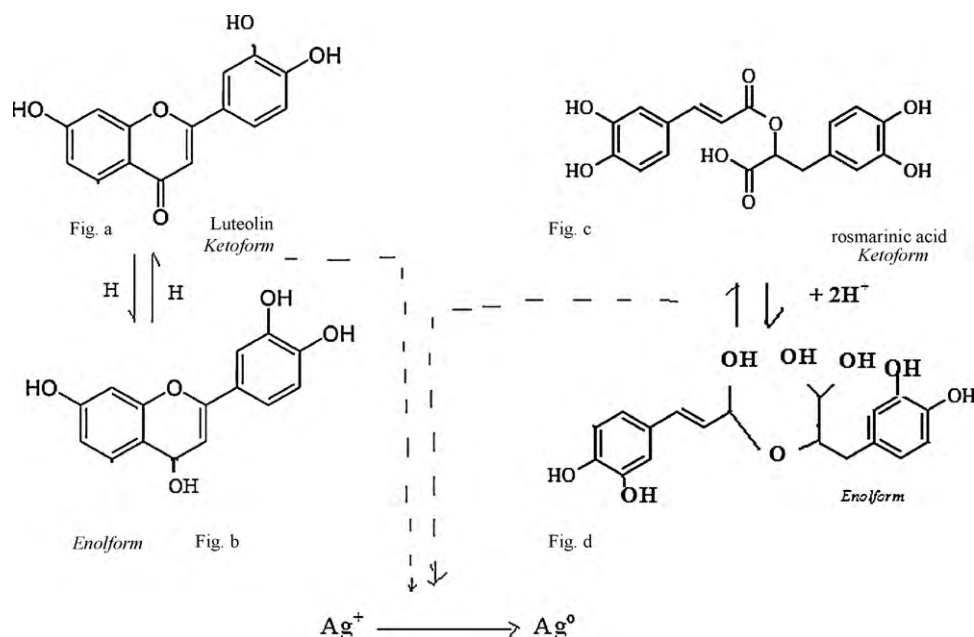


Fig. 6. Mechanism of biosynthesis of Ag nanoparticles.

inflammatory, anti-allergic and antimicrobial capabilities. This acid being an antioxidant, acts also as a free radical scavenger.

Luteolin is a common flavone found in the aerial parts of basil plant. It is a known antioxidant, a free radical scavenger, an agent in the prevention of inflammation and a promoter carbohydrate metabolism.

The formation of enol form of the luteolin freely liberates reactive hydrogen which is responsible for the conversion of Ag⁺ to Ag⁰ (Fig. 6a and b). There is liberation of reactive hydrogen in the keto form of the rosmarinic acid to form an intermediate, unstable enol form of the same (rosmarinic acid). This structure is however unstable due to the presence of two hydroxyl group on one carbon atom and reverts back to its keto form (Fig. 6c and d). Thus in both the cases the reactive hydrogen gets liberated which participates in the synthesis of silver nanoparticles. The antioxidant effectiveness of many polyphenols is essentially a result of the ease with which hydroxyl group is donated to a free radical and the ability of the aromatic structure to support an unpaired electron.

The stability of the silver nanoparticles can be attributed to the formation of silver electride that may form a thin layer on the aqueous surface of the reaction mixture. This silver electride possibly may convert the silver to nanosilver. The protein present further is believed to cap the silver nanoparticles formed, restricting the agglomeration of the particles and thus checking the size and shape.

The exact mechanism of the formation of these nanoparticles in these biological media is unknown. Presumably biosynthetic products or reduced cofactors play an important role in the reduction of respective salts to nanoparticles. It seems quite probable that the phenols play an important part in the reduction of ions to AgNPs as the concept of antioxidant action of phenol compounds is not new.

Nanoparticle application in catalyst, sensors and medicine depends critically on the size and composition of the nanoparticles. Thus different routes leading to the synthesis of nanoparticles of various shapes and sizes have extended the choice of properties that can be obtained. The shape of the nanoparticles also contributes significantly to modulating their electronic properties [31]. The shape of almost all plant mediated AgNPs were mostly spherical with exception of neem (*Azadirachta indica*) which yielded polydisperse particles both with spherical and flat plate like morphology of 5–35 nm in size [32].

AgNPs synthesized from *Cinnamon zeylandicum* leaves extracts ranged at an average size of 31–40 nm with few large size nanoparticles at 100 nm [33]. Magnolia mediated AgNPs ranged from 15 to 500 nm [34] whereas persimmon leaf broth yielded average diameter of 32 nm nanoparticles at 55 °C. Average size with lemon grass extract varies from 50 to 16 nm at 25 °C. TEM images of silver nanoparticles from *Embllica officinaris* were also predominantly spherical with an average size of 16.8 nm ranging from 7.5 to 25 nm [35]. However all these nanoparticles were phase transferred to into an organic solution, i.e. cationic surfactant octadecylamine [31]. In this work, the basil mediated spherical and uniformly sized AgNPs were synthesized at room temperature with only water and no phase transfer was needed in this case.

5. Conclusion

Biosynthesis of stable and spherical shaped nanoparticles of average size ~10 nm has been described using different parts of basil plant. The method is simple and is carried out at room temperature and offers a viable and an ecofriendly way of fabrication of benign nanoparticles without any huge inputs in terms of energy and waste. It is superior to the microbial synthesis as it is synthesized in aqueous solutions at ambient temperature, without any additive or accelerants like ammonia and in lesser time and could be exploited for developing cost effective biosynthesis of Ag nanoparticles at a large scale. Phenolic and flavanoid compounds play an important role in the formation of nanosilver.

Acknowledgements

The authors wish to acknowledge University Grants Commission, New Delhi, India for the financial support under the major research project scheme.

References

- [1] M. Gratzel, Nature 414 (2001) 338.
- [2] M. Okuda, Y. Kobayashi, K. Suzuki, K. Sonoda, T. Kondoh, A. Wagawa, A. Kondo, H. Yoshimura, Nano Lett. 5 (2005) 991.
- [3] J. Dai, M.L. Bruening, Nano Lett. 2 (2002) 497.
- [4] C.B. Murray, S. Sun, H. Doyle, T. Betley, MRS Bull. 26 (2001) 985.

- [5] A. Tao, F. Kim, Y. Sun, Y. Xia, P. Yang, *Nano Lett.* 3 (2003) 1229.
- [6] Y. Shiraiishi, N. Toshima, *J. Mol. Catal. A: Chem.* 141 (1999) 187.
- [7] Y. Sun, Y. Yin, B.T. Mayers, T. Herricks, Y. Xia, *Chem. Mater.* 14 (2002) 4736.
- [8] B. Yin, H. Ma, S. Wang, S. Chen, *J. Phys. Chem. B* 107 (2003) 8898.
- [9] N.M. Dimitrijevic, D.M. Bartels, C.D. Jonah, K. Takahashi, T. Rajh, *J. Phys. Chem. B* 105 (2001) 954.
- [10] A. Callegari, D. Tonti, M. Chergui, *Nano Lett.* 3 (2003) 1565.
- [11] L. Zhang, Y.H. Shen, A.J. Xie, S.K. Li, B.K. Jin, Q.F. Zhang, *J. Phys. Chem. B* 110 (2006) 6615.
- [12] A. Swami, P.R. Selvakannan, R. Pasricha, M. Sastry, *J. Phys. Chem. B* 108 (2004) 19269.
- [13] R.R. Naik, S.J. Stringer, G. Agarwal, S. Jones, M.O. Stone, *Nat. Mater.* 1 (2002) 169.
- [14] M.N. Nadagouda, R.S. Varma, *Green Chem.* 8 (2006) 516.
- [15] P.K. Vemula, U. Aslam, V.A. Mallia, G. John, *Chem. Mater.* 19 (2007) 138.
- [16] J.L. Gardea-Torresdey, E. Gomez, J.R. Peralta-Videa, J.G. Parsons, H. Troiani, M.J. Yacaman, *Langmuir* 19 (2003) 1357.
- [17] D. Rautaray, A. Sanyal, A. Bharde, A. Ahmad, M. Sastry, *Cryst. Growth Des.* 5 (2005) 399.
- [18] S.S. Shankar, A. Rai, A. Ahmad, M. Sastry, *Chem. Mater.* 17 (2005) 566.
- [19] S.S. Shankar, A. Ahmad, M. Sastry, *Biotechnol. Prog.* 19 (2003) 1627.
- [20] T.A. Davis, B. Volesky, R.H.S.F. Vieira, *Water Res.* 34 (2000) 4270.
- [21] D.C. Sharma, C.F. Forster, *Water Res.* 27 (1993) 1201.
- [22] Rajeshwarisivaraj, P. Sivakumar, V. Senthilkumar, Subburam, *Bioresour. Technol.* 80 (2001) 233.
- [23] Y. Nuhoglu, E. Oguz, *Biochemistry* 38 (2003) 1627.
- [24] P. Kumar, P. King, V.S.R.K. Prasad, *J. Hazard. Mater.* B137 (2006) 1211.
- [25] C. Gruen, *Biochim. Biophys. Acta* 368 (1975) 270.
- [26] H. Borchert, E.V. Shevchenko, A. Robert, I. Mekis, A. Kornowski, G. Grubel, H. Weller, *Langmuir* 21 (2005) 1931.
- [27] A. Farukh, I. Ahmad, Z. Mehmood, *Turk. J. Biol.* 30 (2006) 177.
- [28] P. Umadevi, *Indian J. Exp. Biol.* 39 (2001) 185.
- [29] J.B. Harborne, *Biochemistry of Phenolic Compounds*, Academic Press, London, 1964, p. 511.
- [30] J.F. Morgan, R.V. Klucas, R.J. Grayer, J. Abian, M. Becana, *Biol. Med.* 22 (1997) 861.
- [31] V. Kumar, S.K. Yadav, *J. Chem. Technol. Biotechnol.* 84 (2009) 151.
- [32] S.S. Shanker, A. Rai, A. Ahmad, M. Sastry, *J. Colloid Interface Sci.* 275 (2004) 496.
- [33] M. Sathishkumar, K. Sneha, S.W. Won, C.-W. Cho, S. Kim, Y.-S. Yun, *Colloids Surf. B: Biointerfaces* 73 (2009) 332.
- [34] J.Y. Song, B.S. Kim, *Bioprocess Biosyst. Eng.* 22 (2009) 79.
- [35] B. Ankamwar, C. Damle, A. Ahmad, M. Sastry, *J. Nanosci. Nanotechnology* 5 (2005) 1665.